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ORIGINAL ARTICLE

Investigating single-molecule fluorescence quenching and molecular motion dynamics at transparent conductive oxide interfaces

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Abstract
The interaction between single-molecule (SM) fluorescence and transparent conductive oxide interface presents unique opportunities for studying molecular motion dynamics and conformational changes. In this study, we investigate the quenching effect of indium-tin oxide (ITO) on SM fluorescence, focusing on the fluorescent dye Cy3 tethered to the 3′-end of single-stranded DNA (ssDNA). By examining the brightness variations of single Cy3 molecules, we are able to distinguish Cy3-ssDNA covalently attached onto the ITO surface from the case of adsorption. Additionally, we can evaluate the molecular motion dynamics of single ssDNA molecules of varying lengths and conformations on the ITO surface. We believe that our findings make significant contributions to the understanding of molecular interactions at ITO interfaces and offer valuable insights into the potential applications of novel fluorophore motion- and orientation-based biosensing strategies.

Keywords Single-molecule imaging · ssDNA conformation · Fluorescence quenching · Semiconductor interfaces · Nanometer height

1 Introduction

Transparent conductive oxides, including indium-tin oxide (ITO), have been widely used in semiconductor devices, biosensing, imaging, and display applications due to their excellent combination of optical transparency and electrical conductivity [1–5]. Although ITO is optically transparent in the visible range, the molecular fluorescence is usually to be quenched, especially when fluorophores are within a 10^1 nm distance near ITO surfaces [6]. The quenching effect is considered a process of energy transfer from an excited molecule to a nearby semiconductor [7–9]. This results in a decreased quantum yield of single molecules (SMs), leading to a shortened fluorescence lifetime and a reduction in emitted photon counts [10–12]. A decrease in photon counts is usually a disadvantage for the studies that rely on fluorescence intensities. Various approaches have been introduced to increase the distance between the fluorophores and the ITO surface, including the addition of spacers, polymer layers, and silane modification. In contrast, the sensitivity of the fluorescence quenching effect to the fluorophore–ITO distance presents an opportunity for a sensitive biosensing strategy to study molecular interactions and conformational changes, where fluorophores are in close proximity to the ITO interface [13, 14].

Moerland and Hoogenboom [6] established a theoretical model to describe the relationship between the quantum yield of a fluorophore (or an emission dipole) and its height above a planar metal or semiconductor substrate. This model enables precise measurement of axial distance in the range of 1–10 nm above ITO with subnanometer accuracy. In practical applications, the fluorescence lifetime is often used to quantify the modulation in quantum yield. Enderlein and co-workers [15–17] reported a series of methods based on the same principle, metal-induced energy transfer, to resolve the axial localization of SMs and to profile interactions between cellular basal membrane and substrates. Recent work by Tinnefeld and co-workers [18] extended this approach to graphene substrate, enabling the investigation of DNA–protein interactions with spatial resolution down to the Ångström scale.

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An alternative method is to measure the fluorescence intensity, which requires less sophisticated instruments. Lu and Xie [12] quantified experimentally that the fluorescence intensities of fluorophores on ITO surfaces are typically an order of magnitude weaker than the intensities on the coverslips. However, at the SM level, there are intensity variations across different molecules, as well as fluctuations in intensity for the same molecules over time. In addition to the dipole's distance from the ITO surface, variations in detected SM brightness also depend on the dipole orientations with respect to the surface, the stochastic nature of photon emission, and the dynamic behaviors of the fluorophore in its local environment [19, 20].

The stochastic nature of SM emission makes it challenging to attribute brightness fluctuations solely to the quenching effect of the ITO interface, complicating the precise quantification of the SM–ITO distance. However, capturing SMs, resolving their point spread functions (PSFs), and calculating their brightness are more effectively accomplished through SM imaging techniques. These methods offer numerous potential applications, including SM tracking and SM localization microscopy (SMLM) for super-resolution imaging, with recent advancements achieving localization precision down to the Ångström level [21–24]. When extended to spectral or multicolor imaging regimes, these enable single-molecule Förster resonance energy transfer (smFRET)-based studies and multiplexed SM counting and detection applications [25–27]. These approaches facilitate detailed investigations of molecular dynamics and interactions, playing increasingly important roles in SM research.

Herein, we propose a strategy that directly uses the brightness of SMs to evaluate the fluorescent molecular motion dynamics in the axial direction on ITO. The fluorescent molecule, Cy3, is tethered to the 3′-end of a single-stranded DNA (ssDNA) molecule, which is either covalently bound to or adsorbed on the ITO surface. The inherent flexibility of the ssDNA strand and the linker of the Cy3 molecule allows it to rotate and bend freely, enabling it to adopt various conformations. This flexibility prevents the Cy3 molecule from being fixed in a certain orientation, reducing unwanted bias in the measured SM intensities. Additionally, the axial distance between the Cy3 dye and the ITO surface is modulated by the conformational change of the ssDNA molecules. Although we cannot completely eliminate fluctuations in SM intensities caused by factors other than the Cy3–ITO distance, we propose control experiments and data analysis procedures to emphasize that the intensity fluctuations are primarily due to the conformational change in ssDNA, allowing us to evaluate the Cy3–ITO distance in nanometers and to distinguish the molecular motion dynamics of single ssDNA molecules of varying lengths and conformations.

2 Methods

2.1 Materials

The ssDNA samples purified by high-performance liquid chromatography (HPLC) were purchased from Sangon Biotech (Shanghai, China). Their sequences are listed in Table 1. Sodium carbonate (Na_2CO_3), sodium bicarbonate ($NaHCO_3$), (3-glycidyloxypropyl)trimethoxysilane (GOPS), (3-aminopropyl)triethoxysilane (APTES), tris(hydroxymethyl)aminomethane (Tris), Cy3, and ethanol were purchased from Aladdin (Shanghai, China). Acetone and hydrochloric acid (HCl) were purchased from Modern Oriental (Beijing, China). ITO substrates were obtained from SPI Supplies (West Chester, USA) (6488-AB, 15–30 Ω·cm). Glass coverslips were purchased from Paul Marienfeld GmbH (Germany) (Marienfeld Superior, thickness No. 1.5H, tolerance ± 5 μm). Deionized water (> 18 MΩ·cm) was used for all aqueous solutions.

2.2 ssDNA immobilization on the substrates

The glass coverslips (thickness of 170 ± 5 μm) and ITO-coated coverslips were first cleaned sequentially in acetone, ethanol, and H_2O using an ultrasonicator, with each step lasting 15 min. The cleaned substrates were then functionalized with reactive epoxy groups through gas-phase silanization of GOPS for 48 h.

For covalent ssDNA immobilization, two substrates were sandwiched with 15 μL of 20 μM 5′-amino-modified Cy3-ssDNA in H_2O and incubated for 12 h at 37 °C under saturated humidity. The unreacted reactive epoxy groups were quenched by immersing the substrates in a 10 mM Tris buffer (pH 7.7) for 16 h. The prepared ssDNA-modified

Table 1 The list of ssDNA sequences used in this study

DNA name	Sequence
18-mer Cy3-ssDNA	NH_2-$(CH_2)_6$–5′-CCC GGT TGG TGT GGT TGG-3′-Cy3
12-mer Cy3-ssDNA	NH_2-$(CH_2)_6$–5′-CCC GGT TGG TGT-3′-Cy3
6-mer Cy3-ssDNA	NH_2-$(CH_2)_6$–5′-CCC GGT-3′-Cy3
20-mer hairpin Cy3-ssDNA	NH_2-$(CH_2)_6$–5′-<u>CCA AC</u>G GTT GGT GTG <u>GTT GG</u>–3′-Cy3[a]

[a]Underlining marks the complementary sequences forming the hairpin structure

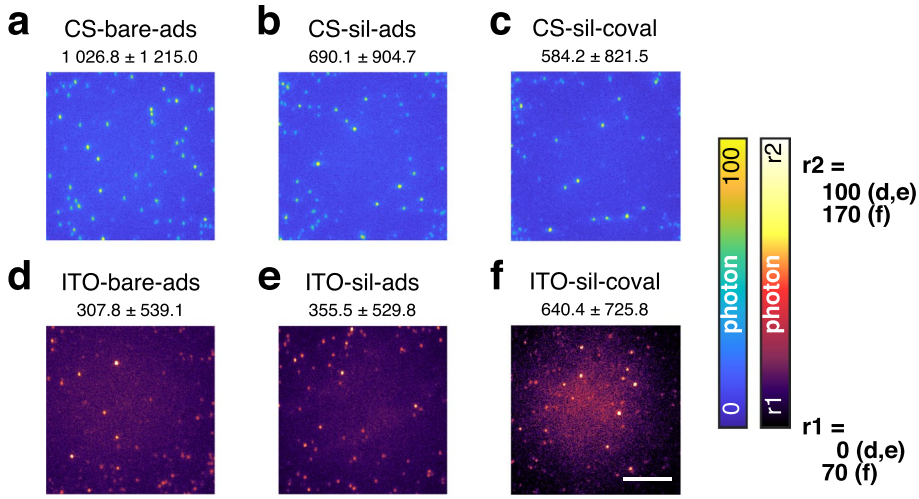

Fig. 1 Representative frames of single Cy3-ssDNA (18-mer) molecule imaging on **a** CS-bare-ads, **b** CS-sil-ads, **c** CS-sil-coval, **d** ITO-bare-ads, **e** ITO-sil-ads, and **f** ITO-sil-coval. Scale bar: 5 µm. The single-molecule brightness is reported as median ± standard deviation (std) in photons, over 6.5×10^3–2.6×10^4 recognized SMs for each case

glass and ITO coverslips were then immediately used for SM fluorescence imaging.

For nonspecific adsorption of Cy3-ssDNA, all reactive epoxy groups of the as-prepared GOPS-treated substrates were immediately quenched using 10 mM Tris buffer (pH 7.7). For comparison, bare substrates without GOPS treatment and epoxy group quenching were also used. Both types of substrates were incubated with 200 µL of 500 nM 5′-amino-modified Cy3-ssDNA in Na_2CO_3–$NaHCO_3$ buffer (10 mM, pH 9.2) for 60 min at room temperature. The excess free Cy3-ssDNA molecules were removed by thoroughly rinsing the substrates with Na_2CO_3–$NaHCO_3$ buffer (10 mM, pH 9.2).

We therefore obtained three types of Cy3-ssDNA-modified surface on two substrates, including Cy3-ssDNA adsorbed on bare glass and ITO coverslips (CS-bare-ads, ITO-bare-ads), Cy3-ssDNA adsorbed on deactivated silane glass and ITO coverslips (CS-sil-ads, ITO-sil-ads), and Cy3-ssDNA covalently bound to silane glass and ITO coverslips (CS-sil-coval, ITO-sil-coval).

2.3 SM fluorescence imaging

The glass and ITO coverslips with covalently modified or adsorbed Cy3-ssDNA were transferred to our home-built microscope equipped with a 100 × objective lens (NA 1.50, UPLAPO100XOHR, Olympus, Japan) for SM fluorescence imaging. For Cy3-ssDNA imaging, a 561-nm laser with a peak intensity of 0.41 $kW\cdot cm^{-2}$ ($1/e^2$ beam diameter) was used. The emission was separated by a dichroic beamsplitter (ZT488/561rpc, Chroma, USA) and filtered using a bandpass filter (ZET488/561m, Chroma, USA). The SM fluorescence images were captured with a typical 50-ms integration time using an sCMOS camera (ORCA-Fusion, Hamamatsu, Japan). All imaging experiments were carried out in Na_2CO_3–$NaHCO_3$ buffer (10 mM, pH 9.2).

ThunderSTORM [28], an ImageJ plugin, was used to detect the PSFs of SMs and resolve their positions (x, y coordinates) along with their sigma values for each frame in the captured raw image stacks. The brightness (in photons) per localization was calculated by summing all photons within a region of interest (11 × 11 pixels) centered at the x and y coordinates of each molecule. The integrated photon count was then background corrected using the average photon count per pixel in the surrounding region. Localizations with sigma values <50 nm or >200 nm or brightness <100 photons were discarded to remove artifacts from diffusing and dim SMs.

3 Results and discussion

Cy3-labeled ssDNA (Cy3-ssDNA, 18-mer) was used as a model system to quantify the quenching behaviors near ITO-coated coverslip surfaces at the SM level. On both substrates with various Cy3-ssDNA modification procedures, single Cy3 molecules were clearly resolved as shown in Fig. 1. Cy3 molecules exhibit similar brightness on the glass coverslips, regardless of the modification procedures used to attach Cy3-ssDNA (see Fig. 1a–c). The detected photons (brightness) for SMs were quantified as 1 026.8 ± 1 215.0 for CS-bare-ads, 690.1 ± 904.7 for CS-sil-ads, and 584.2 ± 821.5 for CS-sil-coval (median ± standard deviation (std)). Direct adsorption of Cy3-ssDNA on ITO results in a dramatic reduction in SM brightness (see Fig. 1d, e, 307.8 ± 539.1 for ITO-bare-ads, 355.5 ± 529.8 for ITO-sil-ads). It is not surprising that ITO is able to effectively quench the emission from Cy3, resulting in dim SMs. The quenching effect is further attenuated while the emitter moves away from the ITO surface. In the case of ITO-sil-coval, as the Cy3 probe was spaced by an 18-mer DNA strand from the ITO surface, its brightness was

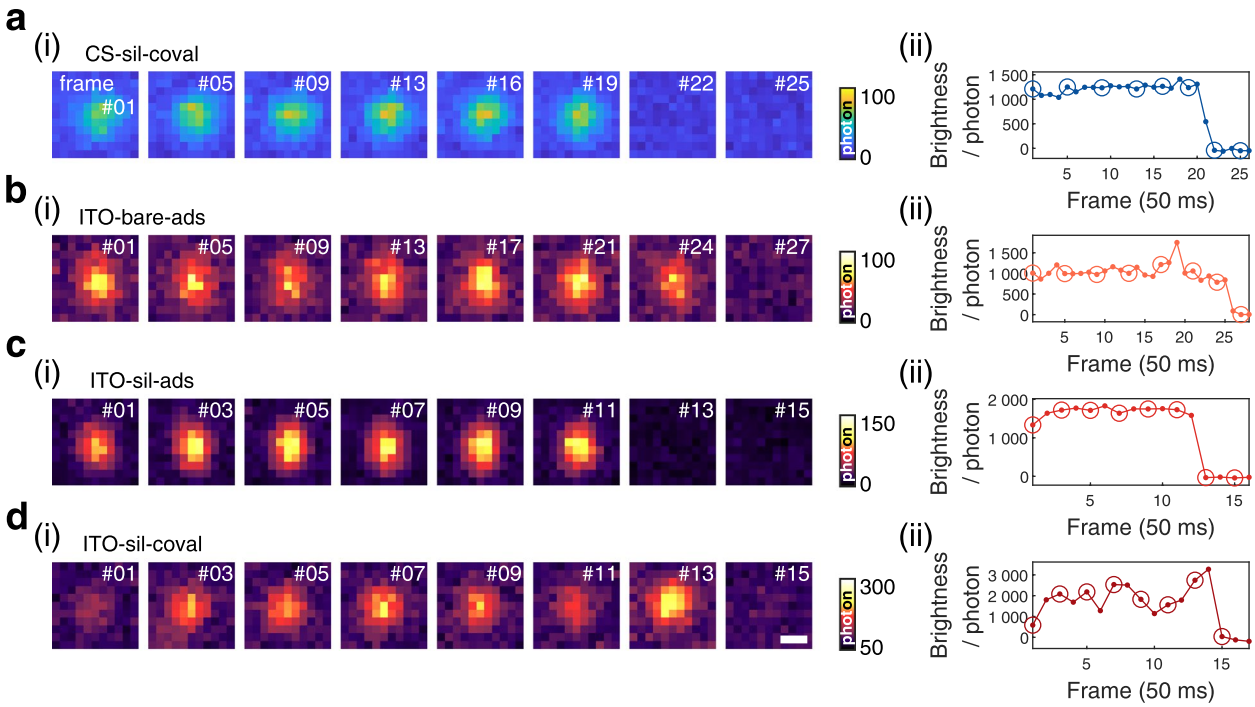

Fig. 2 (i) Snapshots of single Cy3-ssDNA (18-mer) molecule imaging on **a** CS-sil-coval, **b** ITO-bare-ads, **c** ITO-sil-ads, and **d** ITO-sil-coval and (ii) the corresponding trajectories. The frame indices of single-molecule snapshots shown in (i) are highlighted as open circles in (ii). Scale bar: 200 nm

remarkably recovered (640.4 ± 725.8 photons, see Fig. 1f; Table S1 Group 1).

The ability to resolve individual Cy3-ssDNA molecules allows us to characterize their emission behaviors and temporal dynamics at the SM level. We focused on the PSFs whose trajectories exhibited distinctive single-step turn-off patterns, ensuring that we were analyzing individual molecules. For the Cy3-ssDNA molecules on the coverslips, regardless of their immobilization procedures, the brightness trajectories of SMs are more stable over time (see Figs. 2a, S1a, b), indicating the photostability of Cy3 on the coverslip. Similar stable trajectories were observed on both ITO-bare-ads and ITO-sil-ads surfaces (see Fig. 2b, c). However, in the case of ITO-sil-coval, there is a higher probability of observing SMs with fluctuating brightness within single trajectories (see Fig. 2d).

To quantify the brightness fluctuation kinetics of individual Cy3-ssDNA fluorescence across multiple camera frames, we traced the emissive states of individual SMs and grouped the successive frames of each molecule into a burst. The total emitted photons of a burst (for example, N^j_{burst} for the jth burst) were calculated as the sum of the SM localization brightness in the ith frame ($N^{i,j}_{\text{loc}}$) for all k frames until photobleaching occurs in frame $k+1$, so $N^j_{\text{burst}} = \sum_{i=1}^{k} N^{i,j}_{\text{loc}}$. We specifically focused on the bright Cy3-ssDNA bursts lasting multiple frames $\left(N^j_{\text{burst}} \geq 5 \times 10^3 \text{ photons}, k \geq 5\right)$ to exclude short-lived and dim SM bursts.

For all samples of Cy3-ssDNA on various surfaces, we identified the SM traces and calculated $N^{i,j}_{\text{loc}}$ and N^j_{burst} for each trace j. We further introduced $\text{std}(N^{i,j}_{\text{loc}})$, the standard deviation of $N^{i,j}_{\text{loc}}$ in a single trace, and used $\text{std}(N^{i,j}_{\text{loc}})/N^j_{\text{burst}}$ to assess the degree of SM brightness fluctuations within each trace j. On glass coverslip surfaces (CS-bare-ads, CS-sil-ads, CS-sil-coval), we observed no significant variations in N^j_{burst}, $\text{std}(N^{i,j}_{\text{loc}})$, and $\text{std}(N^{i,j}_{\text{loc}})/N^j_{\text{burst}}$ (see Fig. 3a; Table 2). In contrast, the ITO surfaces (ITO-bare-ads, ITO-sil-ads, ITO-sil-coval) exhibited relatively low N^j_{burst} values, consistent with the statistical results for $N^{i,j}_{\text{loc}}$ (see Fig. 1). Notably, the ITO-sil-coval sample displayed the highest $\text{std}(N^{i,j}_{\text{loc}})/N^j_{\text{burst}}$ value (see Fig. 3b (iii) orange plot; Table 2) among the six samples, indicating that it had the most fluctuating SM brightness, as illustrated in Fig. 2a–d (ii).

Quantifying SM brightness at both the localization and burst levels enables us to capture detailed information about molecular motion dynamics at the SM level.

First, it is interesting to note that the same Cy3-ssDNA molecules attached to the ITO surface using different procedures exhibit distinct brightness fluctuation signatures. We believe that the observed fluctuations mostly arise from the varying distance between the fluorophores and the ITO surface, with closer distances resulting in dimmer emission [6]. In the case of adsorption, the cleaned bare ITO (ITO-bare-ads) and Tris-quenched GOPS silane surface (ITO-sil-ads) both hold a large amount of carboxyl and hydroxyl

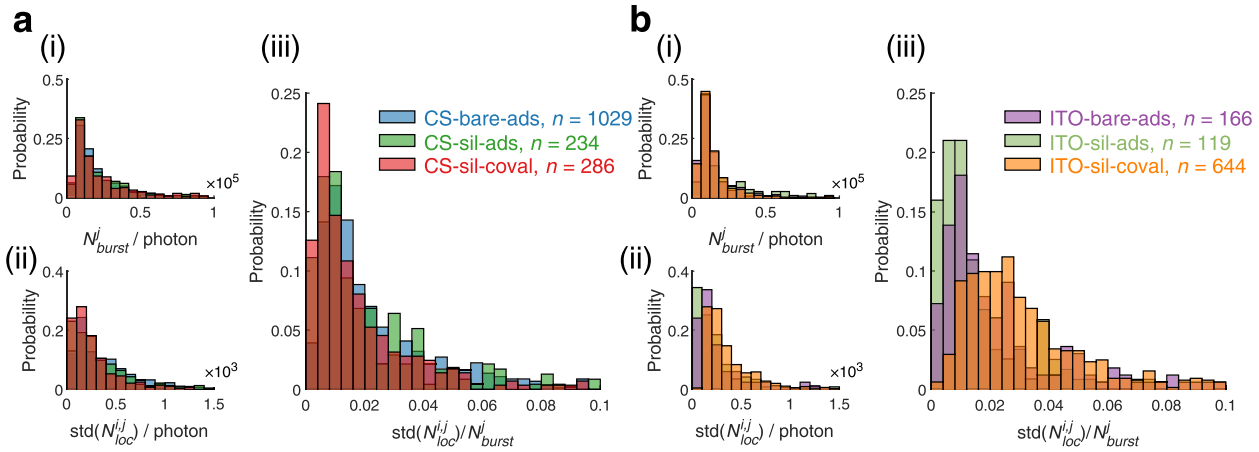

Fig. 3 Histograms of (i) total photons emitted per burst (N^j_{burst}), (ii) standard deviation of detected photons per localization for a burst ($std(N^{i,j}_{loc})$), and (iii) degree of SM brightness fluctuations within each burst ($std(N^{i,j}_{loc})/N^j_{burst}$) on **a** CS-bare-ads, CS-sil-ads, and CS-sil-coval surfaces, and **b** ITO-bare-ads, ITO-sil-ads, and ITO-sil-coval surfaces of 18-mer Cy3-ssDNA. The bursts were screened using the threshold that SMs remain in the emissive state for no less than five successive frames ($k \geq 5$), with a total emitted photon count (N^j_{burst}) exceeding 5 000 before photobleaching occurred. Note that N^j_{burst}, $std(N^{i,j}_{loc})$, and $std(N^{i,j}_{loc})/N^j_{burst}$ were calculated on a per-burst basis, with n representing the total number of bursts recognized from raw image stacks on each surface

Table 2 Calculated values of N^j_{burst}, $std(N^{i,j}_{loc})$, $std(N^{i,j}_{loc})/N^j_{burst}$, and total number of bursts n on CS-bare-ads, CS-sil-ads, CS-sil-coval, ITO-bare-ads, ITO-sil-ads, and ITO-sil-coval surfaces, after filtering with the threshold of $N^j_{burst} \geq 5 \times 10^3$ photons and $k \geq 5$

	$N^j_{burst}(=\sum_{i=1}^{k} N^{i,j}_{loc})$/photon (median ± std)	$std(N^{i,j}_{loc})$/photon (median ± std)	$std(N^{i,j}_{loc})/N^j_{burst}$ **(median ± std)**	Total number of bursts # (n)
CS-bare-ads	15 359.6 ± 22 038.7	301.6 ± 439.6	0.016 3 ± 0.024 9	1 029
CS-sil-ads	15 376.9 ± 30 911.0	226.8 ± 336.3	0.012 7 ± 0.025 6	234
CS-sil-coval	15 125.3 ± 35 911.1	187.9 ± 307.7	0.011 5 ± 0.019 8	286
ITO-bare-ads	9 979.1 ± 22 705.1	170.1 ± 363.2	0.015 6 ± 0.025 6	166
ITO-sil-ads	11 473.3 ± 19 742.2	125.8 ± 218.3	0.010 6 ± 0.016 4	119
ITO-sil-coval	9 929.8 ± 14 128.6	275.0 ± 293.9	0.026 7 ± 0.026 8	644

groups, resulting in negatively charged surfaces [29], where the adsorption of negatively charged Cy3-ssDNA molecule is unfavorable. While Mg^{2+} is commonly used to promote ssDNA adsorption, our control experiment data indicated that Mg^{2+} significantly deteriorates the photostability of Cy3 probes. Therefore, we use an Na_2CO_3–$NaHCO_3$ buffer (10 mM, pH 9.2) as the imaging buffer, as the presence of sodium ions can facilitate the ssDNA adsorption process [30]. The stable brightness trajectories of ITO-bare-ads and ITO-sil-ads reveal the stable adsorption of Cy3-ssDNA on both bare and silane-modified ITO surfaces (see Fig. 2b, c).

Next, on the surface of ITO-sil-ads, although the GOPS silane formed a self-assembled monolayer [31] on the ITO surface and acted as a spacer (approximately 1–2 nm) to increase the distance between Cy3-ssDNA molecules and ITO, it did not promote an increase in SM brightness compared with the Cy3-ssDNA directly adsorbed on the ITO surface (ITO-bare-ads, see Fig. 1d, e). Similar trends were observed for Cy3-ssDNA adsorbed on a Tris-quenched GOPS silane surface compared with a bare coverslip (see Fig. 1a, b) and for Cy3 molecules adsorbed on either coverslip or ITO surfaces (see Fig. S2a; Table S1 Group 3). We hypothesize that the presence of GOPS or Tris moieties in the local environment of Cy3 and Cy3-ssDNA molecules contributes to the decrease in Cy3 SM brightness.

On the contrary, when we replaced the Tris-quenched GOPS silane with APTES silane on the ITO surface, the adsorbed Cy3-ssDNA exhibited a significant increase in brightness compared with the Cy3-ssDNA adsorbed on bare ITO (651.2 ± 989.6 for ITO-sil(APTES)-ads and 307.8 ± 539.1 for ITO-bare-ads (see Table S1 Group 2; Figs. S2b, 1d, e). This trend indicates that brighter Cy3-ssDNA is observed when it is positioned further from the ITO surface. However, the epoxy group of GOPS is necessary for covalently conjugating 5′-amino-modified Cy3-ssDNA so that covalent immobilization experiments had to be conducted on GOPS silane-modified substrates.

Finally, we specifically focused on the case of 18-mer Cy3-ssDNA covalently conjugated to the ITO substrates (ITO-sil-coval). In this scenario, the 3′-end of Cy3 is able to move to a height of approximately 8 nm above the ITO surface in the extended conformation of the 18-mer ssDNA

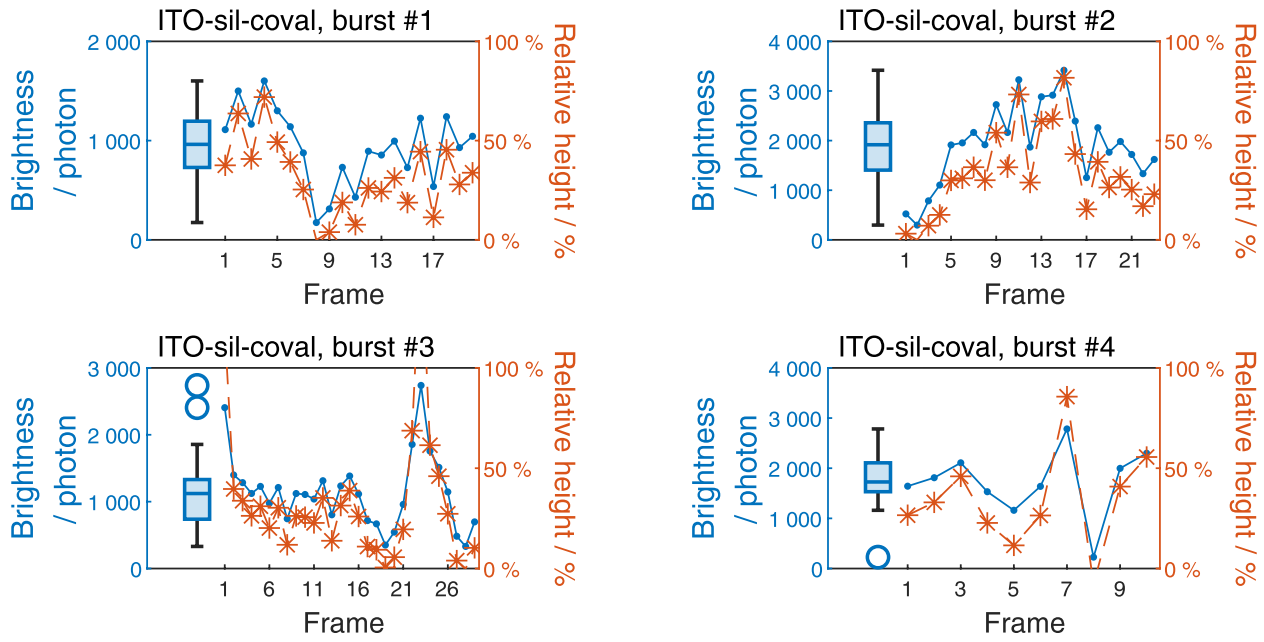

Fig. 4 Boxplots of SM brightness of 18-mer Cy3-ssDNA within a single trajectory (blue profiles). The SM brightness was converted to relative height above the ITO-sil-coval surface (orange profiles) in the 2–8-nm range (see Supplementary Note 1)

strand in the axial direction or move as close as possible to approximately 2 nm above the ITO surface when the ssDNA strand leans on the silane surface (see Supplementary Note 1). The latter case results in similar Cy3 brightness as observed in ITO-sil-ads. To extract height information from variations in SM brightness, we conducted further analysis based on the following assumptions:

(i) We have demonstrated that ITO-sil-coval exhibits the largest SM brightness variations in SM trajectories compared with all adsorption cases on ITO and all cases on glass coverslips (see Figs. 2a–d (ii) and 3b (iii)). This suggests that the brightness fluctuations inherent to Cy3 are significantly smaller than the height-induced quenching effect from ITO. In other words, the brightness fluctuations of SM Cy3-ssDNA primarily originated from the conformational changes of the ssDNA tethered to the ITO surface.

(ii) The 18-mer ssDNA sequence used in this study does not exhibit favorable secondary structures analyzed by NUPACK [32], indicating that the ssDNA is more likely to undergo stochastic conformational changes driven by thermal fluctuations. We assume that Cy3 has equal probabilities of occupying any position within its maximum and minimum height range, provided we have sufficient sampling of the SM brightness.

We therefore focused on longer-lasting SM bursts. Within each burst trace j, the SM brightness of each localization $N_{\mathrm{loc}}^{i,j}$ can be treated as a limited sampling of brightness fluctuations. A boxplot (see Fig. 4, left panels) quantifies the maximum and minimum values, as well as the outliers of $N_{\mathrm{loc}}^{i,j}$ in the burst trace j, based on the 1.5 interquartile range (IQR) criterion [33]. The maximum and minimum values of $N_{\mathrm{loc}}^{i,j}$ are correlated to the extended conformation of the ssDNA strand in the axial direction and to the ssDNA strand settled in a leaning position on the silane surface, which is approximately within the 2–8-nm range (see Supplementary Note 1) above the ITO surface. Within this range, the quantum yield recovers almost linearly as the dipole height increases logarithmically [6]. Based on this mathematical relation, we were able to convert the SM brightness $N_{\mathrm{loc}}^{i,j}$ to the relative height above the silane surface on the ITO substrate (see Figs. 4, S4, right panels; Supplementary Note 2).

In bursts #3 and #4 of Fig. 4, lower and upper bound outliers were detected in some frames, resulting in estimated Cy3 heights <0% and >100% of relative height. We believe that these outliers are more likely attributed to brightness fluctuations from the Cy3 fluorophore rather than to the height-induced quenching effects related to ITO.

Using the data and analysis framework established above, we further extended our strategy to measure various ssDNA molecules of different lengths and conformations. By decreasing the number of nucleotides in Cy3-ssDNA covalently attached to the ITO surface, from 18-mer to 6-mer, we hypothesized that the Cy3-tethered 6-mer ssDNA would be

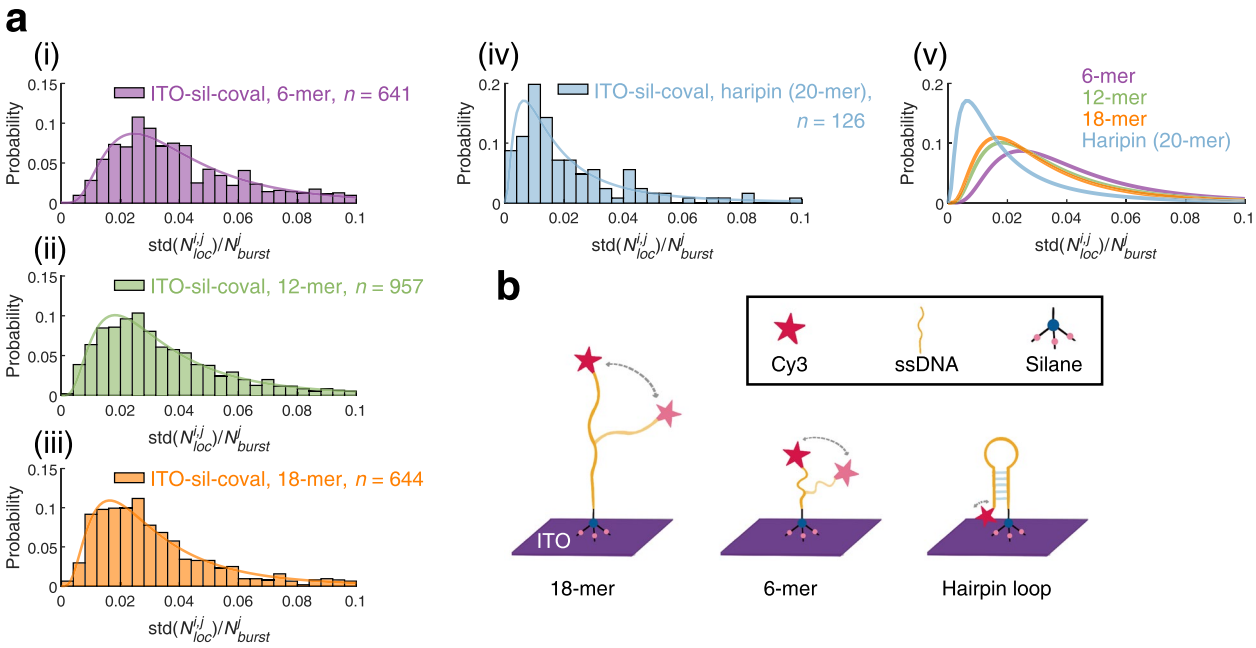

Fig. 5 **a** Histograms of SM brightness fluctuations within each burst ($\mathrm{std}(N^{i,j}_{\mathrm{loc}})/N^{j}_{\mathrm{burst}}$) of (i) 6-mer Cy3-ssDNA, (ii) 12-mer Cy3-ssDNA, (iii) 18-mer Cy3-ssDNA, and (iv) 20-mer hairpin Cy3-ssDNA on ITO-sil-coval surfaces. The bursts were screened using the threshold that SMs remain in the emissive state for no less than five successive frames ($k \geq 5$), with a total emitted photon count (N^{j}_{burst}) exceeding 5 000 before photobleaching occurred. Note that $\mathrm{std}(N^{i,j}_{\mathrm{loc}})/N^{j}_{\mathrm{burst}}$ was calculated on a per-burst basis, with N representing the total number of bursts recognized from raw image stacks on each surface. Solid profiles in (i)–(iv) represent lognormal distribution fitted to each distribution and summarized in (v). **b** Schematic illustrating how the height of Cy3 above ITO and its motion dynamics are modulated by the ssDNA lengths and their conformational changes

closest to the ITO surface and thus exhibit a higher quenching probability. However, we observed only an insignificant decrease in both SM brightness and the total photons emitted per burst (N^{j}_{burst}) (see Tables S2 and S3; Fig. S3).

In contrast, the Cy3-tethered 6-mer ssDNA exhibited a dramatic increase in SM brightness fluctuations within each burst (std $(N^{i,j}_{\mathrm{loc}})/N^{j}_{\mathrm{burst}}$) (see Fig. 5a (i)–(iii)). These fluctuations followed a lognormal distribution (see Fig. 5a (v)), and the fitted median ± std and mode ± uncertainty values aligned well with experimental observations (see Table 3).

Although the 18-mer, 12-mer, and 6-mer ssDNA molecules progressively approach the ITO surface, their brightness fluctuations increase nonlinearly. Considering that the quantum yield decreases linearly as the dipole height decreases logarithmically, we conducted simulations to further confirm the experimental observation that a slight height disturbance of the Cy3-tethered 6-mer ssDNA leads to greater fluctuations in Cy3 brightness of 12-mer and 18-mer ssDNA (see Supplementary Note 3).

In a separate experiment, we imaged another ssDNA sequence that preferentially forms a hairpin loop conformation (see Table 1, 20-mer hairpin Cy3-ssDNA). This configuration forces the tethered Cy3 probe to orient much closer to the ITO surfaces, resulting in noticeably dimmer Cy3 signals (see Table S2). Unlike the 6-mer case, we observed significantly reduced SM brightness fluctuations (see Fig. 5a (iv)). The formation of the hairpin loop greatly enhances the rigidity of the stem-loop structure, thereby restricting the intrinsic dynamic motion of the tethered Cy3 probe (see Fig. 5b).

Table 3 Calculated values and fitting results of $\mathrm{std}(N^{i,j}_{\mathrm{loc}})/N^{j}_{\mathrm{burst}}$ of various ssDNA sequences on ITO-sil-coval surface, after filtering with the threshold of $N^{j}_{\mathrm{burst}} \geq 5 \times 10^{3}$ photons and $k \geq 5$

ssDNA sequences	$\mathrm{std}(N^{i,j}_{\mathrm{loc}})/N^{j}_{\mathrm{burst}}$			Total number of bursts # (n)
	Median ± std (experiment)	Median ± std (fitting)	Mode ± uncertainty (fitting)	
18-mer Cy3-ssDNA	0.026 7 ± 0.026 8	0.026 7 ± 0.027 1	0.016 3 ± 0.000 6	644
12-mer Cy3-ssDNA	0.029 1 ± 0.027 0	0.029 2 ± 0.028 7	0.018 2 ± 0.000 6	957
6-mer Cy3-ssDNA	0.035 2 ± 0.028 2	0.036 0 ± 0.029 7	0.024 6 ± 0.000 8	641
20-mer hairpin Cy3-ssDNA	0.014 9 ± 0.023 7	0.015 4 ± 0.028 3	0.006 4 ± 0.000 9	126

4 Conclusions

In this study, we demonstrated the significant influence of the ITO quenching effect on fluorescence intensities at the SM level and validated its potential application in distinguishing SM conformational changes. Our findings indicate that the brightness fluctuations of single Cy3 fluorophores tethered to an ssDNA molecule immobilized on the ITO surface primarily result from the conformational changes in the ssDNA. Through detailed analysis of SM brightness trajectories, we established a method to estimate the height of Cy3 above the ITO substrate, enhancing our understanding of molecular dynamics influenced by semiconductor quenching effects. Additionally, we successfully extended this strategy to image ssDNA molecules of varying lengths and conformations. These insights not only advance the field of SM imaging but also hold potential for developing biosensing technologies based on SM motion, conformation, and orientation.

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Authors' contributions Jin Lu conceived and designed the experiments. Zihan Pan performed the experiments. Jin Lu and Zihan Pan analyzed the data and wrote the paper. Jin Lu approved the final version of the manuscript.

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Data availability All data generated or analyzed during this study are included in this published article and its supplementary information files.

Declarations

Ethics approval and consent to participate Not applicable.

Consent for publication All authors have approved the manuscript and agree to its publication in this journal.

Competing interests The authors declare no competing financial interest.

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